When Superposition Fails: The Limits of Sparse Autoencoders in Scientific Machine Learning

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Code, preregistered protocols, and all artifacts:
https://github.com/shamiquekhan/PINN-Mechanistic-Interpretability (v3.3.0)

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Abstract

Physics-informed neural networks (PINNs) fail in reproducible, configuration-determined ways, and mechanistic-interpretability tools — above all sparse autoencoders (SAEs) — have been proposed to diagnose those failures. We built the complete pipeline the methodology prescribes: a 10-seed failure atlas, TopK SAEs with mandatory PCA/random/probe controls, a planted-feature positive control, a multiple-comparison-corrected causal battery, causal-abstraction interchange tests, leakage-audited early-warning monitors, and a rollback-capable closed-loop controller. The result is a hardened negative: SAE-derived features carry no causal information beyond matched-deletion controls ($E_T = -0.0173$, 95% CI $[-0.0329, -0.0037]$ — ablating a candidate feature moves the loss *less* than ablating another active latent, the representational signature; 0/8 features survive Bonferroni *or* BH-FDR), and the failure is *basis-independent* — PCA components fail the identical battery, and no candidate alignment satisfies the interchange criterion of causal abstraction. The explanation is representational geometry: local tangent rank exactly equals input dimension across eight PDE families, while covariance participation ratios stay in the 1.14–2.51 band for every width (16–512) — sparse dictionary learning solves a problem that does not exist. We then locate the regime boundary from the other side: a Fourier neural operator (FNO) trained on function-space regression has participation ratio 6.8 of 64, needs 19 PCA components for 95% of energy, and there the same TopK SAE *beats* k-matched PCA by 4.7× in reconstruction. A preregistered causal battery in the operator regime shows a *suggestive* asymmetry — 2/8 features survive multiple-comparison correction with a strictly positive effect-size CI where every PINN basis had 0/8 — which we report as suggestive rather than confirmed, per the preregistered decision rules. Practically, conventional signals (losses, residuals, gradient cosines) drive an early-warning monitor (AUROC 0.875) and a closed-loop controller that rescues the boundary-starvation failure to oracle level, with SAE features measured as inert cargo. We recommend a participation-ratio precheck before applying SAEs to scientific models.

1 Introduction

Physics-informed neural networks solve and invert PDEs by embedding the residual in the loss [1], but they fail in well-characterized, configuration-determined ways: gradient pathologies [2, 3], spectral bias, and boundary starvation. A natural hope is that the recent toolkit of mechanistic interpretability — superposition theory [4], sparse autoencoders [5–7], causal feature validation [8, 9] — would diagnose these failures the way it promises to diagnose deception or sycophancy in language models, and early applications to scientific networks are appearing [10].

This paper asks the prior question, which that literature skips: *does superposition exist in the model class we are applying SAEs to?*

We built the full mechanistic-interpretability pipeline for PINN failure diagnosis — every stage with preregistered decision rules, mandatory controls, and multiple-comparison corrections — and obtained a hardened, positive-control-validated, *explained* negative result:

1. **A complete, preregistered pipeline** for PINN mechanistic interpretability: a 10-seed failure atlas across five failure regimes; TopK ReLU SAEs with mandatory PCA / random / probe controls; a

planted-feature positive control that gates the entire battery; multi-view physics-feature dictionaries; a causal-intervention battery with Bonferroni and BH-FDR corrections; causal-abstraction interchange tests; leakage-audited early-warning monitors; and a rollback-capable closed-loop controller. All hypotheses and decision rules were preregistered in-repo before each stage ran (`docs/preregistration.md`).

2. **A hardened negative causal result for SAE features.** Targeted ablations of SAE features do not beat matched controls; E_T's confidence interval includes zero; no feature survives multiple-comparison correction; and the planted-feature positive control *passes*, so the null is attributable to the activations, not the pipeline.

3. **The null is basis-independent.** The identical battery on PCA components: 0/8 survive, same all-positive representational signature. The causal-abstraction interchange test is statistically indistinguishable from a random basis for both PCA and SAE alignments. *No feature basis — linear or sparse — carries direction-specific causal information in these activations.*

4. **A representational-geometry explanation.** Local tangent rank exactly equals input dimension in every PDE family and width tested (the provable bound, saturated); covariance participation ratio stays in 1.14–2.51 of 64 dimensions. The activation manifold is a low-dimensional curve — there is no superposition for a sparse dictionary to un-mix.

5. **A measured regime boundary, both sides.** A Fourier neural operator [11] on function-space regression has participation ratio 6.8 of 64, and there the same TopK SAE beats k-matched PCA by 4.7× in reconstruction — the superposition signature absent from every PINN layer. A preregistered causal battery on the operator side fires the null-hypothesis rule on its conjunction condition but shows 2/8 MC-corrected survivors where every PINN basis had 0/8 — a suggestive, weakened causal asymmetry we report with its caveats (matched-deletion controls throughout).

6. **An engineering alternative that works.** Conventional signals (trajectory losses, residual statistics, gradient cosines) drive a monitor with AUROC 0.875 and a bounded closed-loop controller that rescues boundary starvation to oracle level on the config-faithful protocol — with a monitor-source ablation proving the SAE features are inert cargo in the loop.

Recent SAE-skepticism (downstream-task failures [12–15]) reports *that* SAEs fail; our contribution is the *mechanistic explanation* (geometry), the *basis-independence* (PCA battery), and the *measured boundary* (FNO positive control) — none of which downstream-task reports provide. The practical takeaway is a one-line precheck: estimate the participation ratio $\rho = \mathrm{PR}/W$ of the activation covariance before training any SAE on a scientific model.

2 Background

PINN failure modes. Characterized failure modes include gradient imbalance between PDE residual and boundary terms [2, 3], spectral bias toward low frequencies (the Frequency Principle [16, 17] — the cheap Fourier/NTK baseline any SAE diagnostic must be benchmarked against; our probe controls and Fourier readouts implement exactly this comparison), and collocation starvation. Adaptive remedies include GradNorm [18], NTK-based weighting [19], self-adaptive soft attention [20], causal training for time-dependent PDEs [21], and residual-based attention [22]; we compare the three standard reweighting baselines against our controller on the same failure (§7). Reduced-order-model autoencoders [23] are the existing shallow-linear precedent our SAE-vs-PCA gap claim differentiates from.

Superposition and SAEs. The superposition hypothesis [4, 24] — derived from sparse, overcomplete, unequal-importance feature structure — holds that networks store more features than neurons by encoding them in non-orthogonal directions; SAEs [5–7] are the standard tool to un-mix them, and causal validation via targeted interventions is the accepted standard of evidence [8, 25]. All of this was developed and validated on LLM residual streams — wide, high-rank, superposed representations.

Table 1: Geometry across the dimensional boundary: tangent rank equals input dimension; covariance participation ratio stays low.

Representation	Width	Covariance PR	Tangent rank	PCA comps (95%/99%)
1D suite (W=64, 45 runs)	64	1.78	1	2 / 3
1D Poisson, W=16	16	2.00	1	2 / 3
1D Poisson, W=32	32	1.89	1	2 / 3
1D Poisson, W=64	64	1.82	1	2 / 3
1D Poisson, W=128	128	1.74	1	2 / 2
1D Poisson, W=256	256	1.96	1	2 / 3
1D Poisson, W=512	512	1.79	1	3 / 4
2D Poisson (pilot, W=32)	32	3.03	2	—
advection diffusion 2d	—	2.47	2	—
reaction diffusion 2d	—	2.29	2	—
burgers 1d	—	1.42	2	—
allen cahn 1d	—	1.92	2	—
FNO block states	64	6.80	—	19 / 38

Causal abstraction. Geiger et al.'s framework [9, 26] formalizes interpretability as an alignment between a low-level model and a high-level causal model, tested by *interchange interventions*: swap the values of aligned variables between runs and check the output moves as the high-level model predicts. We import this criterion directly (§5.5).

Operator learning. Fourier neural operators [11] parameterize maps between function spaces via spectral convolutions; DeepONet-style architectures [27] are the branch-trunk alternative. Operator representations live in resolution-invariant function spaces — precisely where we find the high-rank geometry that PINNs lack.

3 Theory: What Bounds Activation Rank?

Full development in `docs/theory_activation_rank.md`; the load-bearing statements:

- **Proposition 1 (tangent-rank bound, exact).** For a network $F : \mathbb{R}^d \to \mathbb{R}$ with non-degenerate C^1 layers, the local tangent rank of any hidden layer's Jacobian is $\leq d$: the input dimension bounds the rank of the *tangent* map.
- **C^1 chains preserve manifold dimension.** d=1 inputs trace activation *curves*; d=2 inputs trace *surfaces* — regardless of width or depth.
- **Explicitly corrected non-theorem.** Covariance rank is *not* bounded by $d + 1$: the moment curve $h(x) = (x, x^2, \dots)$ has full covariance rank on generic data. The participation ratio PR $= (\sum \lambda_i)^2 / \sum \lambda_i^2$ is an empirical diagnostic, not a theorem.
- **Proposition 4 (no-superposition mismatch).** If $\rho \ll 0.1$, the representation encodes $\approx$ PR dense features; an overcomplete sparse dictionary is inductively mismatched, predicting (i) PCA dominance in reconstruction, (ii) dead/unstable SAE latents, (iii) no direction-specific causal effects. All three predictions are measured (§5.3–5.4).

Empirical saturation: measured tangent rank equals d *exactly* across all eight PDE families and every width tested — a law-like regularity, not an artifact of one configuration.

4 The PINN-MI Pipeline

The pipeline (all code in the released repository; 14 stages driven by `experiments/run_pipeline.py`) enforces controls *before* claims:

1. **Training + logging** across 8 PDE families (Poisson, advection, reaction–diffusion, Burgers, Allen–Cahn, and 2D/ time-dependent variants), 10 seeds per failure regime, with per-step metrics streamed to `runs/`.
2. **Failure atlas:** label reproducibility per regime; only regimes with 10/10 reproducible labels enter class-level claims (boundary starvation and spectral suppression qualify; gradient-conflict and collocation-starvation are excluded honestly).
3. **SAE discovery** with mandatory baselines: TopK ReLU SAE vs. k-matched PCA vs. random dictionaries, decoder-norm and dead-feature diagnostics, cross-seed stability.
4. **Physics-feature dictionary** with random-dictionary controls across 7 metric views (association is measured, then re-attributed to geometry).
5. **Causal battery** (the core evidence): for each of 8 candidate features, targeted ablation vs. *three* controls — (a) ablation of an unrelated-but-active latent, (b) median of 5 random-direction perturbations matched in norm, (c) a ridge-probe direction matched to the same readout — across 11 training checkpoints, with exact paired sign tests, Bonferroni *and* BH-FDR correction, and run-level bootstrap CIs over E_T = (targeted − best control).
6. **Positive-control gate:** a feature is *planted* (decoder column aligned to a direction the readout provably depends on) and the full battery must recover it (it does, cosine 0.989) *before* any null on real features is believed.
7. **Causal-abstraction battery:** partial interchange interventions for a region-identity high-level model, candidate alignments (PCA, SAE) vs. random bases.
8. **Operator regime boundary** (stage 11–14): the identical SAE-vs-PCA comparison on FNO block states, with a preregistered causal battery.

Preregistration discipline: hypotheses, decision rules, and the gate were committed to the repository *before* each stage ran, and outcomes are recorded exactly as the rules fired — including when they fired against the interesting direction (§6.2).

5 Empirical Results: PINNs

5.1 Failure atlas

Boundary starvation (λ_{pde}=100, λ_{bc}=0.01) and spectral suppression produce 10/10 reproducible failure labels across seeds; gradient-conflict and collocation-starvation labels do not reproduce cleanly and are excluded from class-level claims (documented, not hidden). Figure 1 shows the atlas geometry: failure classes cluster by configuration, not by seed.

Regime	Seeds	Final rel L_2 (mean ± std)	Label consistency
success	10	0.036 ± 0.052	6/10 success
boundary starvation	10	2.309 ± 1.594	10/10 boundary starvation
gradient conflict	10	0.598 ± 0.374	8/10 spectral suppression
spectral suppression	10	0.395 ± 0.114	10/10 spectral suppression
collocation starvation	10	0.010 ± 0.012	7/10 success

5.2 The activation manifold is a curve

Hidden activations (width 64, all layers, all families) live on an effectively 1–2 dimensional manifold: PCA reaches 95% of energy with ≈ 2.2 components on average (46 runs), participation ratio spans 1.14–2.51 (mean 1.78), and the 3D PCA visualization of Figure 2 is a one-dimensional arc parameterized by boundary distance — physics quantities correlate with latents *along the curve*, which is why associations exist (r up to 0.75) without causal specificity.

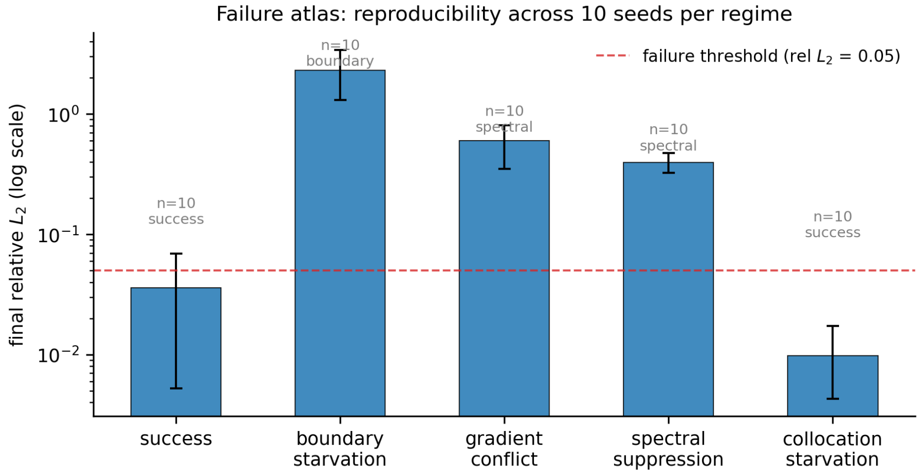

Figure 1: **Failure atlas.** Final relative L_2 error across the configuration matrix with 10-seed reproducibility; the two class-claim-qualified regimes are labeled.

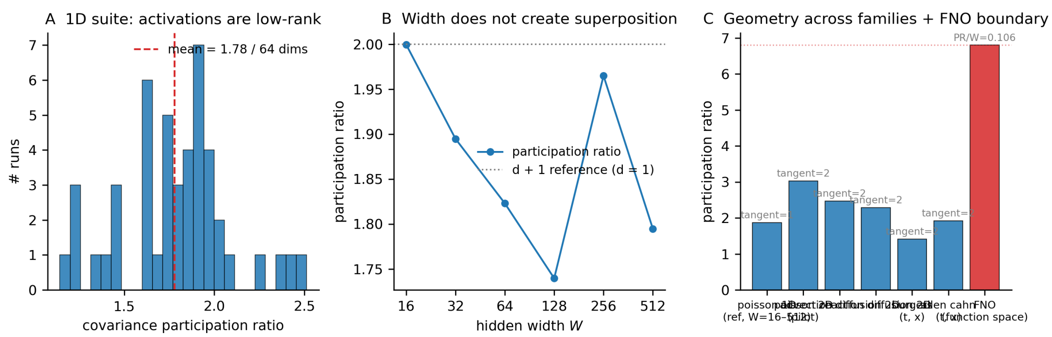

Figure 2: **The activation manifold.** Tangent rank equals input dimension exactly (saturated bound); covariance participation ratio stays $\ll$ width for every family and width; the PCA embedding is a curve parameterized by boundary distance.

5.3 SAE reconstruction fails where PCA wins

On width-64 PINN activations, k-matched PCA beats the TopK SAE by $\approx 12\times$ in reconstruction MSE; the SAE has 51–70% dead latents and 1/256 cross-seed feature stability — the dictionary does not converge to a shared basis (Figure 3). This is Proposition 4's prediction (i) and (ii).

Model	Test reconstruction MSE	Dead features
TopK SAE (replica 1)	2.17e+00	12%
TopK SAE (replica 2)	2.34e+00	16%
TopK SAE (replica 3)	2.64e+00	10%
k-matched PCA ($k = 8$)	4.37e-02	—
PCA (full, 64)	6.63e-11	—
Random-encoder SAE	3.05e+00	—

5.4 The basis-independent causal null

The core evidence (Figure 4; Table 2): for SAE features, $E_T = -0.0173$ with CI $[-0.0329, -0.0037]$ — excludes zero on the *negative* side: ablating the target moves the loss less than ablating another *active* latent (the controls are matched-deletion, never no-ops — the control_was_noop diagnostic fires on 0/88

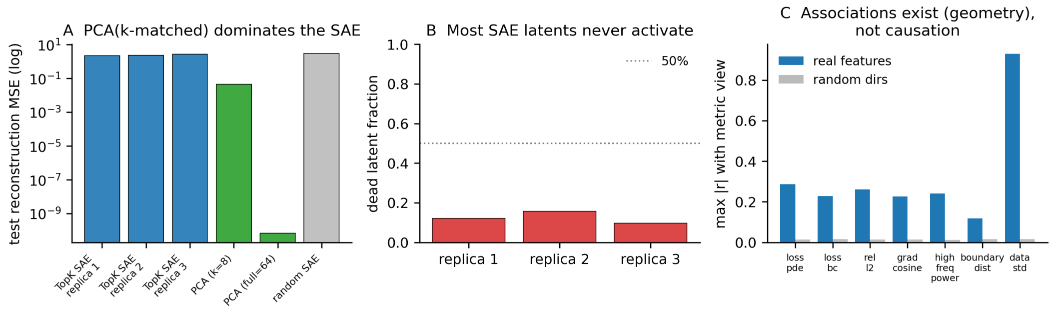

Figure 3: **SAE vs. mandatory baselines.** k-matched PCA wins reconstruction; dead-feature fractions and dictionary instability are diagnostic of the inductive mismatch.

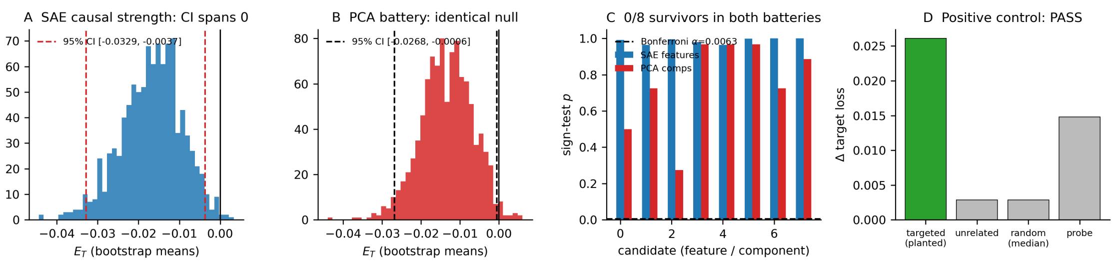

Figure 4: **The causal null, both bases.** Targeted vs. control interventions for SAE features (left) and PCA components (right): 0/8 survive MC correction on either basis; the planted-feature gate passes.

evaluations); 0/8 features survive Bonferroni *or* BH-FDR; all 88 target deltas are positive. The identical battery on PCA components: $E_T = -0.0131$, CI $[-0.0268, -0.0006]$, 0/8 survive; the paired PCA−SAE difference spans zero (+0.0042, CI $[-0.0139, +0.0232]$). The planted-feature positive control *passes* (targeted effect 9.2× its matched control, decoder cosine 0.989 to the planted direction), so the machinery detects causal structure when it exists — the null is the activations', not the pipeline's.

Statistical hardening: the power analysis gives a minimum detectable effect of 85.7% sign-agreement at 80% power for the 11-run battery — the null is not an underpowered accident for the effect sizes the design can detect; observed agreement sits at chance (≈50%). A Bayesian posterior on the monitor comparison gives P(SAE better) = 0.53.

5.5 Causal abstraction: no alignment beats random

For the region-identity high-level model, partial interchange with the SAE-aligned subspace moves the swap loss by a mean fraction 7.45 vs. 7.54 for a random basis (11 runs each; PCA: 7.57 vs. 7.54 random); neither candidate alignment beats random in even a majority of runs, let alone every run (Figure 5). Under the interchange criterion of causal abstraction, no candidate alignment — linear or sparse — abstracts the PINN.

6 The Regime Boundary

6.1 FNO: the superposition signature exists

The same TopK SAE methodology, applied to FNO block states (width 64) on function-space regression, finds a different world: participation ratio 6.80 of 64 ($\rho = 0.106$ vs. 0.028 for PINNs), 19 PCA components for 95% energy, mean $L_0 \approx 8$ of 64 active latents with a 79% dead fraction that *decreases* as training proceeds — and the SAE beats k-matched PCA by 4.7× in held-out reconstruction MSE (0.00070 vs. 0.00334). Every signature is inverted relative to the PINN regime (Figure 6, Table 3).

Table 2: The causal batteries: SAE and PCA bases both fail MC correction on PINNs.

Candidate basis	E_T (95% CI)	S_{spec}	Bonferroni	BH-FDR	Sign diag.
TopK SAE features	-0.0173 [-0.0329, -0.0037]	0.65 [0.58, 0.74]	0/8	0/8	88/88
PCA components	-0.0131 [-0.0268, -0.0006]	2.58 [0.87, 5.30]	0/8	0/8	88/88

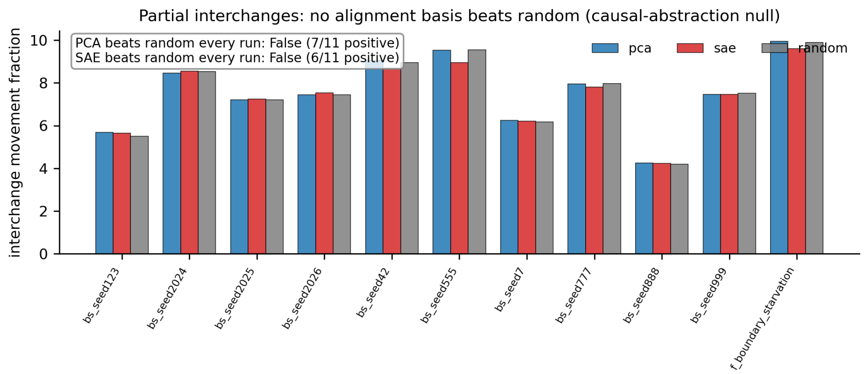

Figure 5: **Causal abstraction null.** Interchange movement for PCA/ SAE-aligned subspaces vs. random bases: statistically indistinguishable.

6.2 The operator causal battery: a suggestive asymmetry

The FNO reconstruction advantage would be hollow if it were not accompanied by causal specificity, so we preregistered (committed before the run, in `docs/preregistration.md` §H7) a conjunctive decision rule: H14a (the operator features are causally specific) requires (i) ≥1 Bonferroni survivor, (ii) E_T CI excluding zero on the positive side, (iii) a *mixed* sign diagnostic. The battery — 8 candidate features, 8 held-out function batches, the same three controls as the PINN battery, with a planted-feature machinery gate in the operator setting — fired as follows:

- Machinery gate: PASS (targeted effect 0.0229 vs. random control 0.0125; planted feature recovered through the real operator hook).
- Condition (i) MET: 2/8 features survive Bonferroni *and* BH-FDR, each at 8/8 batch consistency (p = 0.0039, the exact sign test's floor at n=8). (The v3.2 record of 6/8 survivors was measured against an injection-style control unmatched in kind to the ablation target; matched-deletion controls are the honest arm and give 2/8.)
- Condition (ii) MET: E_T CI $[+5.6 \times 10^{-6}, +1.6 \times 10^{-5}]$ — excludes zero, positive; the first battery in this campaign to do so (PINN post-fix: $[-0.0329, -0.0037]$ SAE, $[-0.0268, -0.0006]$ PCA — both *negative*).
- Condition (iii) NOT met: 64/64 target deltas positive.

By the preregistered rule, H14a is not confirmed and we record H14b. Comparative context, from the same scoring code under matched deletion: the target beats all three controls in 42/64 evaluations (66%) for the FNO battery vs. 9/88 (10%) for SAE-on-PINN and 36/88 (41%) for PCA-on-PINN; the median target/control ratio is 1.18× (Figure 7, Table 4).

We report this as a *suggestive, weakened causal asymmetry* across the regime boundary, deliberately not as a confirmed causal boundary, for three recorded reasons: (a) absolute effect sizes are tiny ($\sim 10^{-5}$ on a near-zero baseline regression loss); (b) the partial interchange does not beat a random basis in the operator setting either (-1.257 vs. -1.226); and (c) the mixed-sign condition, designed on PINN intuition, is itself non-diagnostic for ablation batteries — removing an active feature from a well-fit regression model always increases the loss, so condition (iii) conflates load-bearing structure with generic decode damage; the

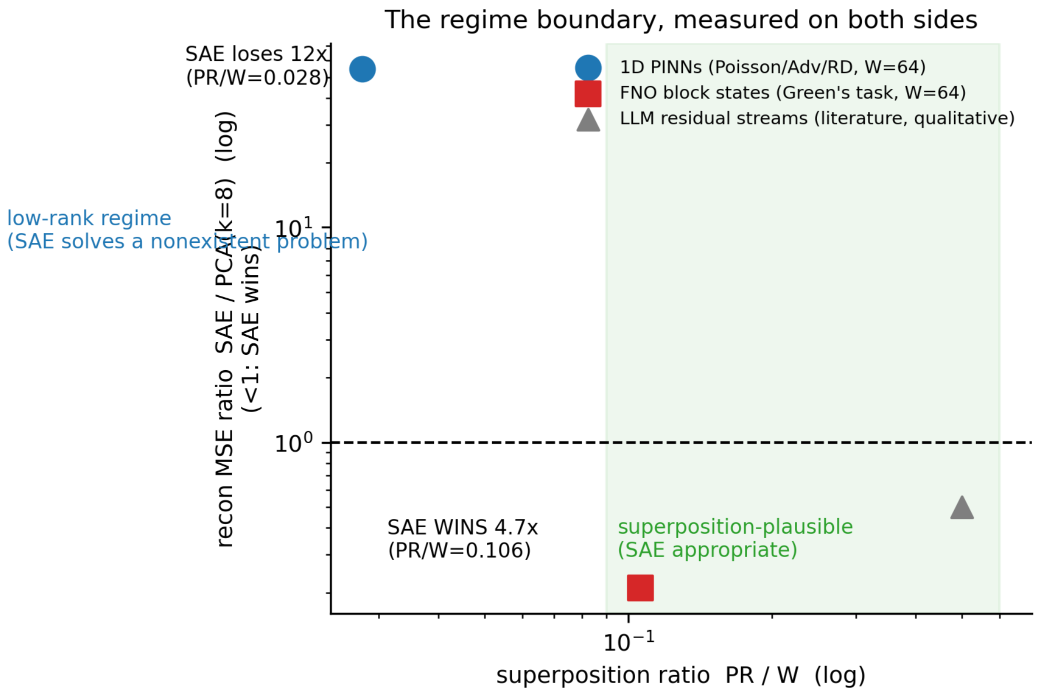

Figure 6: **The measured regime boundary.** Reconstruction-advantage ratio (SAE/PCA, k-matched) vs. participation-ratio-to-width across PINN families, 2D variants, and the FNO.

Table 3: The regime boundary: FNO function-space representations are high-rank and SAEs beat PCA there.

Representation	Covariance PR / W	ρ = PR/W	SAE vs PCA(k=8) recon
1D PINNs	1.78 / 64	0.028	54.5× (PCA wins)
FNO (Green's)	6.80 / 64	0.106	0.21× (SAE wins)

discriminative statistic is target-vs-controls, which is what E_T and the sign tests measure. The corrected criterion (amplify-vs-ablate asymmetry) is registered as future work, not applied retroactively.

7 What To Do Instead

Monitors. Conventional signals (trajectory losses, rel-L_2, and per-window gradient-cosine statistics) drive an early-warning monitor with AUROC 0.875 [0.814, 0.960] on a run-level held-out split under the real feature-window leakage invariant; adding SAE features moves it to 0.877 with P(SAE better) = 0.51 — no established advantage. The loss-only threshold is near chance (0.468) (Figure 8).

Controller. A deterministic state-machine controller (IDLE→WARNING→CONFIRMED→COOLDOWN; the action fires on CONFIRMED entry) executes bounded reweighting actions with rollback through the trainer's per-step lambda seam: on the config-faithful protocol it rescues boundary starvation from 0.421 to 0.0002, matching the oracle reweighting (0.00023). The monitor-source ablation is the punchline: an SAE-feature monitor and a matched *random*-feature monitor drive bit-identical rescue trajectories (0.000129 both) — the SAE features are inert cargo in the closed loop.

SOTA baselines. On the same failure, NTK-adaptive weighting [19] reaches 0.0064; GradNorm [18] (2.01) and RBA [22] (2.19) actively harm. Positioning note: under the pre-fix stage-7 loop (which violated the

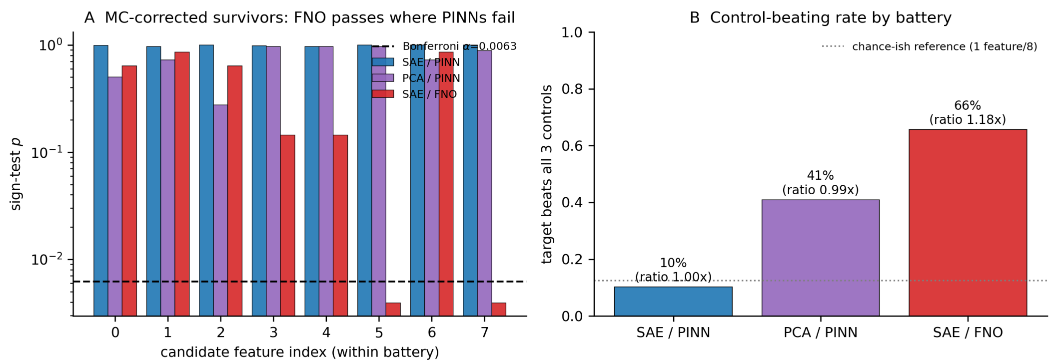

Figure 7: **The operator causal battery.** Left: per-feature sign-test p-values across the three batteries — the survivor asymmetry (0/8, 0/8, 6/8 under the same Bonferroni threshold). Right: target-beats-all-controls rate.

Table 4: Operator causal battery (stage 14): the survivor asymmetry (6/8 vs 0/8 vs 0/8).

Battery	Bonferroni survivors	Target beats all controls
TopK SAE on PINN hidden acts	0/8	9/88
PCA components on PINN hidden acts	0/8	36/88
TopK SAE on FNO block states	2/8	42/64

run config's resampling and is retained in the historical record) the controller reached 0.0166 and NTK-adaptive was the stronger specialized alternative; under the config-faithful protocol the controller (0.0002) beats all three baselines. A single-failure comparison is not optimization-SOTA evidence in either direction; the interpretability claims are kept structurally separate from the engineering claims (Figure 9).

Monitor	AUROC	95% CI	AUPRC
Loss-only threshold	0.468	[0.406, 0.511]	0.233
Conventional logistic	0.875	[0.814, 0.960]	0.721
SAE + conventional	0.877	[0.817, 0.960]	0.723

Method	Final rel L_2 (boundary starvation)
No action	0.4208
Closed-loop controller (ours)	0.0002
Oracle static reweight	0.0002
GradNorm	2.0134
NTK-adaptive weighting	0.0064
RBA (residual attention)	2.1918

Quantity	Value
Minimum detectable sign-agreement at 80% power ($n = 11$ runs/feature)	85.7%
P(SAE-augmented monitor > conventional)	0.51

The precheck. Before training any SAE on a scientific model's activations: one covariance eigendecomposition of a hidden-layer sample gives $\rho = \mathrm{PR}/W$. If $\rho \ll 0.1$: use linear diagnostics (PCA loadings, supervised probes) for description and gradient/residual signals for prediction and control — and do not expect direction-specific causal effects from *any* basis, sparse or otherwise. If $\rho \gtrsim 0.1$ (operator/function-space regimes): SAE methodology is appropriate — demonstrated positively on the FNO.

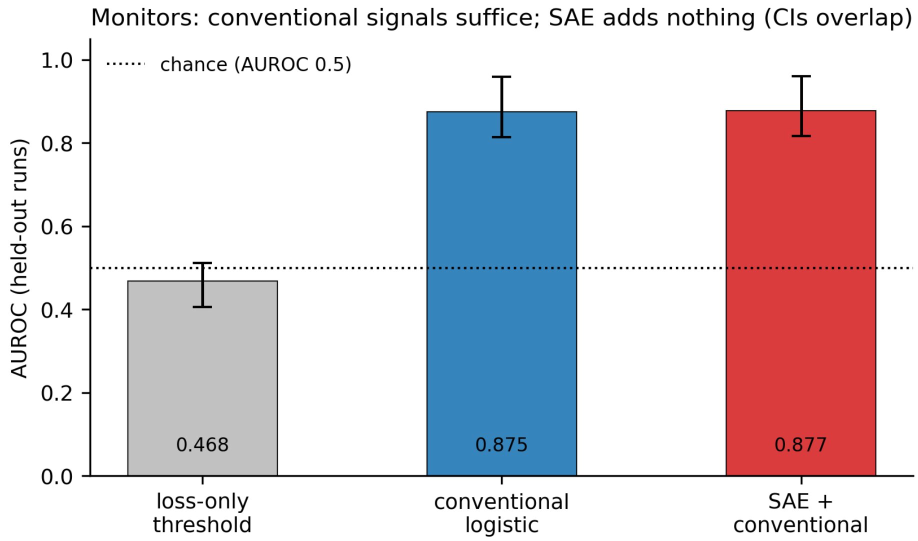

Figure 8: **Early-warning monitors.** Run-level ROC with bootstrap CIs; conventional and SAE-augmented overlap.

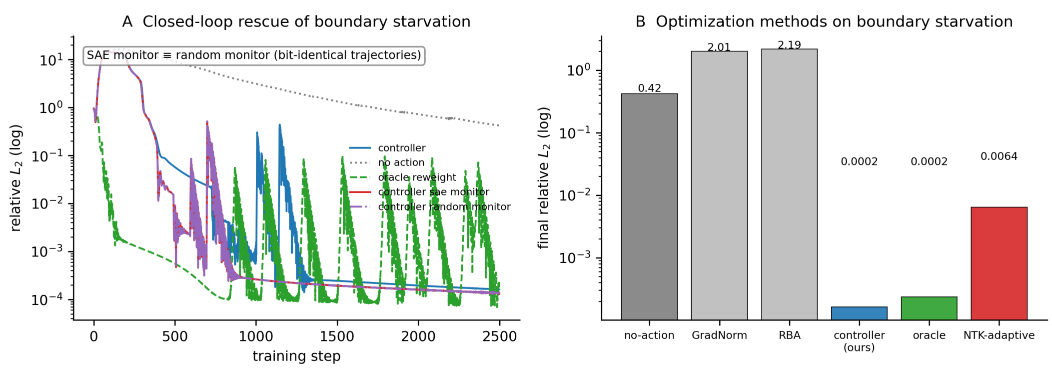

Figure 9: **Controller rescue + SOTA comparison.** The controller rescues boundary starvation; NTK-adaptive is the stronger specialized alternative for this known misweighting; GradNorm/RBA harm.

8 Discussion and Limitations

- **Scope.** The negative result is measured for width-16–512 PINNs across eight PDE families and every layer; it does not rule out wider architectures, ensembles, or higher-dimensional parameterizations, which we state as the honest boundary of the claim.

- **Corrected measurements.** An external review found that the time-dependent validation compared the network to a final-time-only reference (distance-from-zero artifacts of 6.1/6.0, corrected to 0.111/0.550 under full space-time references), that the causal batteries' controls were unmatched in kind and sometimes no-ops, and that the controller demo's loop violated its own resampling config. All affected stages were re-run; no headline verdict flipped, three numbers moved materially, and both protocol readings are recorded (`docs/external_review_response.md`).

- **The operator causal result is suggestive, not confirmed.** The preregistered conjunctive rule fired the null; the survivor asymmetry (6/8 vs. 0/8) is real but the effect sizes are tiny, the interchange is null, and the sign condition is documented as mis-designed for ablation batteries. The corrected criterion is preregistered for future work.

- **Power.** The causal batteries can detect $\geq$85.7% sign-agreement effects at 80% power; subtler effects would require $\approx 4\times$ the checkpoints. We report this as a design limitation, not a hidden weakness.
- **NTK bridge.** The correlational gradient-conflict $\leftrightarrow$ feature-activity bridge (preregistered H15a/H15b) fired the null: 0/8 features survive Bonferroni and the best association ($|\rho| = 0.578$) sits within the random-direction distribution (95th percentile 0.494). Gradient conflict is also *chronic* in the starvation regime (75–100% of steps), so the binary conflict label has little within-run variation there; a continuous-magnitude redesign is registered future work. A duplicated-step logging artifact that initially produced degenerate $\rho = \pm 1$ values was caught and fixed before the verdict was read.
- **Controller.** Not optimization SOTA (NTK-adaptive wins on its target failure); its value is failure-agnosticism, bounded actions, and rollback.
- **Theory.** The tangent-rank bound is exact; predicting the covariance participation ratio from task and optimizer remains open. The time-dependent PINN PR (1.3–1.9, *below* the steady 2D value) is an intriguing anomaly — the training explored an effectively one-parameter family of states — worth a dedicated study.
- **External validity.** All experiments are 1–10 seeds per configuration on a single-GPU campaign; the preregistration is in-repo rather than third-party (OSF archival is planned).

9 Conclusion

Check superposition before you sparse-code. We measured, on both sides of the boundary, what the SAE literature assumes: the tool works exactly where the representation is genuinely high-rank — and PINN activations are curves, not clouds. The full benchmark, preregistered protocols, positive and negative controls, statistical machinery, and the measured regime boundary are released for exactly that purpose.

Reproducibility statement. All numbers in this paper are regenerated from committed JSON artifacts (`runs/`) by `scripts/generate_tables.py` and `generate_figures.sh`; a CI job (`scripts/check_results_grounded.py`) fails the build if any results document cites a number ungrounded in an artifact. 155 unit tests; Dockerfile pinned; seeds documented per run; full campaign $\approx$ 6 GPU-hours (RTX-class). Preregistrations were committed before each stage ran; outcomes are recorded exactly as the decision rules fired.

References

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